NARWCDB Validation Reference

This document is the canonical reference for every data quality rule enforced by the NARWCDB system. Rules are enforced at two layers:

1. Schema constraints — enforced by the SQL Server database at insert/update time. These are absolute hard stops; the database will reject any row that violates them.
2. MATLAB validation rules — run before upload by `narwc.validation.SurveyValidator`. These catch problems in the incoming CSV/data that the schema alone cannot express (range checks, cross-field logic, survey-type-specific rules).

At the end of this document is a complete SAS-to-new-system mapping showing which legacy SAS quality checks have been ported and which are still gaps.

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1. Schema-Level Constraints

Schema constraints are enforced unconditionally by SQL Server. They require no MATLAB code and cannot be bypassed by the uploader.

1.1 NOT NULL columns

Only two columns in the Master table are declared NOT NULL (aside from the surrogate primary key):

Column	Constraint	Notes
`Master_ID`	`int NOT NULL IDENTITY`	Auto-generated surrogate PK; never supplied by submitter
`FILEID`	`varchar(20) NOT NULL`	Every row must identify the survey file it belongs to
`EVENTNO`	`int NOT NULL`	Every row must have a sequential event number within its file

All other columns are nullable. Required-field checks beyond FILEID and EVENTNO are handled by MATLAB validation rather than schema constraints, because requirements are survey-type-dependent.

1.2 Primary key

```
CONSTRAINT PK_Master PRIMARY KEY CLUSTERED (Master_ID ASC)
```

(FILEID, EVENTNO) is not enforced as unique at the database level because legitimate surveys can produce multiple rows sharing the same EVENTNO (e.g., a position event and a sighting event recorded at the same moment). Duplicate detection is handled by MATLAB validation before upload.

1.3 Data type constraints

The column types implicitly enforce range and format:

Column	Type	Implicit constraint
YEAR	smallint	Integer; range −32,768 to 32,767
MONTH	tinyint	Integer; range 0–255
DAY	tinyint	Integer; range 0–255
TIME	int	Integer; no fractional seconds
S_TIME	int	Integer
LAT_DD	decimal(10,5)	Max 5 decimal places; rejects non-numeric
LONG_DD	decimal(10,5)	Max 5 decimal places
S_LAT	decimal(10,5)	Max 5 decimal places
S_LONG	decimal(10,5)	Max 5 decimal places
ALT	decimal(8,2)	Numeric; rejects text
HEADING	smallint	Integer
LEGNO	smallint	Integer
ANGLEL	smallint	Integer
ANGLER	smallint	Integer
VISIBLTY	decimal(8,2)	Numeric
SURFTEMP	decimal(8,2)	Numeric
NUMBER	int	Integer; no fractional counts
NUMCALF	int	Integer

1.4 Foreign key constraints

35 FK constraints link Master columns to lookup table primary keys. A row will be rejected if it supplies a non-NULL value that does not exist in the referenced lookup table. NULL values are always allowed (FK constraints do not apply to NULLs).

Master column	References	Lookup description
ANHEAD	ANHEAD(Value)	Animal heading code (0–15 compass octants, plus codes 21/22)
BEAUFORT	Beaufort(Value)	Beaufort wind/sea-state scale
BEHAV1–BEHAV15	Behave(Value)	Behavior codes (15 FK constraints, one per slot)

Master column	References	Lookup description
`BLOCK`	`Block(Value)`	CCS aerial survey block code
`CLOUD`	`Cloud(Value)`	Cloud cover (oktas)
`CONFIDNC`	`Confidnc(Value)`	Group-size estimate confidence level
`DDSOURCE`	`DDSOURCE(Value)`	Decimal-degree position source
`GLAREL`	`GLARE(Value)`	Sun glare severity, left side
`GLARER`	`GLARE(Value)`	Sun glare severity, right side
`IDREL`	`IDREL(Value)`	Species identification reliability
`IDSOURCE`	`IDSOURCE(Value)`	Species identification source
`LEGSTAGE`	`LEGSTAGE(Value)`	Leg stage within a survey leg
`LEGTYPE`	`LEGTYPE(Value)`	Leg type (transit, census, etc.)
`MONTH`	`[MONTH](Value)`	Calendar month (1–12) or season code (13–16)
`PHOTOS`	`PHOTOS(Value)`	Photo/video documentation codes
`PLATFORM`	`PLATFORM(Value)`	Aircraft or vessel identifier
`SPECCODE`	`SPECCODE(Value)`	Species code (up to 8 characters)
`STRATUM`	`STRATUM(Value)`	Survey stratum code
`STRIP`	`STRIP(Value)`	Distance-interval strip number
`TAXCODE`	`TAXCODE(Value)`	Taxonomic group code
`WX`	`WX(Value)`	Weather code

Note on BEAUFORT: The `Beaufort` lookup table is the authority for which Beaufort values are valid. The legacy SAS system used 0–7 with 9 as a "missing" sentinel; the current lookup table definition should be consulted to confirm the exact valid set.

Note on CLOUD: The `Cloud` lookup table covers the valid cloud cover codes. There is a known discrepancy between the legacy SAS system (which flagged values 5–8 as invalid, using 0–4 and 9) and the current lookup table; domain expert review is needed to resolve this before finalizing the Cloud.csv contents.

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2. MATLAB Validation Rules

MATLAB validation runs before any data reaches the database. Each rule is a function in `src/+narwc/+validation/+rules/` with signature `fn(data,collector,config)`. Rules add findings to an `ErrorCollector` with severity `error` or `warning`. Only records that pass all error-severity checks are uploaded.

2.1 **`required_fields.m`** — Required field presence

Rule ID prefix: `required_fields`

Checks that a configurable list of fields is not NULL or empty. The list is supplied via `config.required_fields`. The default list (from `get_config('validation')`) is the authoritative set; the stub `default_config()` inside the file is marked as inaccurate and should not be relied upon.

Check	Severity	Rule ID
Required column is entirely absent from the data table	error	`required_fields.column_missing`
Required field value is NULL/NaN/empty for a row	error	`required_fields.value_missing`

Notes: The schema enforces NOT NULL on FILEID and EVENTNO directly. This rule extends the NOT NULL check to additional fields that are logically required for a given survey type (e.g., YEAR, MONTH, DAY, LAT_DD, LONG_DD) but are nullable in the schema because requirements vary by survey type.

2.2 **`datetime_rules.m`** — Date and time validation

Rule ID prefix: `datetime_rules`

Validates YEAR, MONTH, DAY, and TIME fields individually and as a calendar date.

YEAR

Check	Severity	Rule ID	Threshold
YEAR outside configured range	error	`datetime_rules.year_out_of_range`	Default: 1900 to current year + 1 (`validation.datetime.year_min/year_max`)
YEAR before warning threshold	warning	`datetime_rules.year_too_old`	Default: before 1980 (`validation.datetime.year_warning`)

MONTH

Check	Severity	Rule ID	Threshold
MONTH outside 1–16	error	`datetime_rules.month_out_of_range`	1–12 = calendar months; 13–16 = season codes

Note: MONTH is also enforced by the FK constraint to the `[MONTH]` lookup table, which covers values 1–16.

DAY

Check	Severity	Rule ID	Threshold
DAY outside 1–31	error	`datetime_rules.day_out_of_range`	Simple numeric bounds

TIME

Check	Severity	Rule ID	Threshold
TIME outside 0–239999	error	`datetime_rules.time_out_of_range`	Must represent a valid HHMMSS integer
TIME hours component ≥ 24	error	`datetime_rules.time_invalid_hours`	Extracted as floor(TIME/10000)
TIME minutes component ≥ 60	error	`datetime_rules.time_invalid_minutes`	Extracted as floor(mod(TIME,10000)/100)
TIME seconds component ≥ 60	error	`datetime_rules.time_invalid_seconds`	Extracted as mod(TIME,100)

Calendar date combination

Check	Severity	Rule ID	Notes
YEAR/MONTH/DAY combination is not a valid calendar date	error	`datetime_rules.invalid_date_combination`	Uses MATLAB datetime() to catch e.g. Feb 30; skipped for season-code months (13–16)

2.3 **`coordinate_rules.m`** — Position validation

Rule ID prefix: `coordinate_rules`

Validates LAT_DD and LONG_DD (decimal degrees). The schema stores these as `decimal(10,5)`, which enforces numeric type; this rule checks logical bounds.

Latitude

Check	Severity	Rule ID	Threshold
LAT_DD is missing	error	`coordinate_rules.lat_missing`	—
LAT_DD outside global range	error	`coordinate_rules.lat_out_of_range`	Default: −90 to 90
LAT_DD outside typical survey area	warning	`coordinate_rules.outside_survey_lat`	Default: 20 to 55 °N (`validation.coordinates.study_area_`

Longitude

Check	Severity	Rule ID	Threshold
LONG_DD is missing	error	`coordinate_rules.lon_missing`	—
LONG_DD outside global range	error	`coordinate_rules.lon_out_of_range`	Default: −180 to 180
LONG_DD outside typical survey area	warning	`coordinate_rules.outside_survey_lon`	Default: −85 to −40 °W (`validation.coordinates.study_area_l`

Coordinate pair consistency

Check	Severity	Rule ID	Notes
One of LAT_DD/LONG_DD present but not the other	error	`coordinate_rules.coordinate_pair_mismatch`	Both must be present or both absent

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2.4 **`beaufort_rules.m`** — Beaufort sea state

Rule ID prefix: `beaufort_rules`

Validates BEAUFORT against the configured set of valid values. The default valid set is 0–12. Only non-NULL values are checked.

Check	Severity	Rule ID	Threshold
BEAUFORT value not in valid set	error	`beaufort_rules.beaufort_out_of_range`	Default: 0–12, configurable via `config.valid_values`

Note: The schema FK constraint on BEAUFORT references the `Beaufort` lookup table and provides a redundant hard stop. This MATLAB rule catches the problem before upload with a more informative message.

2.5 **environmental_rules.m** — Environmental conditions

Rule ID prefix: `environmental_rules`

Validates VISIBLTY and SURFTEMP.

Visibility

Check	Severity	Rule ID	Threshold
VISIBLTY is negative	error	`environmental_rules.visibility_negative`	Applies when `config.visibility_allow_negative` is false
VISIBLTY unusually high	warning	`environmental_rules.visibility_too_high`	Default: > 50 nautical miles

Sea surface temperature

Check	Severity	Rule ID	Threshold
SURFTEMP below ocean minimum	warning	`environmental_rules.surftemp_too_cold`	Default: < −2 °C
SURFTEMP above ocean maximum	warning	`environmental_rules.surftemp_too_hot`	Default: > 35 °C

2.6 **foreign_key_rules.m** — Lookup-table validation

Rule ID prefix: `foreign_key_rules`

Validates 20 fields against their respective lookup tables loaded from `data/tables/`. This is a general-purpose engine: it reads the lookup table, extracts the `Value` column, and reports any non-NULL field values that are not in the valid set.

Fields and their lookup tables:

Field	Lookup table	Notes
`ANHEAD`	`anhead`	Animal heading direction code
`BLOCK`	`block`	CCS survey block; see note on missing code MB

Field	Lookup table	Notes
`CLOUD`	`cloud`	Cloud cover; see CLOUD scale discrepancy note above
`CONFIDNC`	`confidnc`	Group-size confidence (0–11)
`CONTRIB`	`contrib`	Contributing organization
`DDSOURCE`	`ddsource`	Position source
`DTYPE`	`dtype`	Record type (legacy reference table; not FK'd in schema)
`GLAREL`	`glare`	Left-side glare severity (0–3)
`GLARER`	`glare`	Right-side glare severity (0–3)
`IDREL`	`idrel`	Identification reliability
`IDSOURCE`	`idsource`	Identification source
`LEGGOOD`	`leggood`	Leg quality flag (legacy reference; not FK'd in schema)
`LEGSTAGE`	`legstage`	Leg stage code
`LEGTYPE`	`legtype`	Leg type code
`OLDVIZ`	`oldviz`	Legacy visibility code (not FK'd in schema)
`PHOTOS`	`photos`	Photo/video documentation (1–5)
`PLATFORM`	`platform`	Aircraft or vessel identifier
`STRATUM`	`stratum`	Survey stratum
`STRIP`	`strip`	Distance-interval strip number
`WX`	`wx`	Weather code

For each field, the rule flags rows where the value is non-NULL and not present in the lookup table.

Rule ID pattern: `foreign_key_rules.<fieldname>_invalid` (lower-cased field name)

2.7 **`behavioral_rules.m`** — Behavior code validation

Rule ID prefix: `behavioral_rules`

Validates BEHAV1–BEHAV15 codes against the `Behave.csv` lookup table and checks for logical inconsistencies among recorded behaviors.

Code validity

Check	Severity	Rule ID	Notes
BEHAVn value not in Behave.csv	error	`behavioral_rules.invalid_behavior_code`	Applies to all 15 BEHAV slots independently

Note: The schema FK constraints enforce the same rule at the database level for all 15 slots (FK_Master_BEHAV1 through FK_Master_BEHAV15). The MATLAB check runs first and provides row-level detail.

Behavior compatibility

Check	Severity	Rule ID	Notes
Dead/stranded behavior combined with active swimming behavior	error	`behavioral_rules.incompatible_behaviors`	Configured via `config.dead_behaviors` and `config.active_swimming_behaviors`
Declared incompatible behavior pair both present	error	`behavioral_rules.incompatible_behaviors`	Configured via `config.incompatible_behavior_pairs`

Taxcode-behavior compatibility

Check	Severity	Rule ID	Notes
Behavior code not valid for the recorded TAXCODE	warning	`behavioral_rules.taxcode_behavior_restriction`	Driven by `config.taxcode_behavior_restrictions`; no restrictions configured by default

Species-behavior compatibility

Check	Severity	Rule ID	Notes
Behavior code not typical for the recorded SPECCODE	warning	`behavioral_rules.species_behavior_restriction`	Driven by `config.species_behavior_restrictions`; no restrictions configured by default

Calf behavior consistency

Check	Severity	Rule ID	Notes
Calf-associated behavior recorded but NUMCALF is 0 or absent	warning	`behavioral_rules.calf_behavior_no_calf`	Configured via `config.calf_associated_behaviors`; empty by default

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2.8 **`species_rules.m`** — Species and sighting field validation

Rule ID prefix: `species_rules`

Validates SPECCODE, TAXCODE, NUMBER, NUMCALF, and cross-field consistency among them.

SPECCODE

Check	Severity	Rule ID	Notes
SPECCODE has unexpected data type	error	`species_rules.speccode_wrong_type`	Must be char/string
SPECCODE exceeds 4 characters	error	`species_rules.speccode_too_long`	Max 4 chars per domain standard
SPECCODE missing on a sighting record	error	`species_rules.speccode_missing_for_sighting`	Only applies when `config.require_speccode_for_sightings` is true
SPECCODE not found in SPECCODE.csv lookup	error	`species_rules.speccode_not_in_table`	Requires `config.validate_speccode_lookup` = true
SPECCODE contains invalid characters	error	`species_rules.speccode_invalid_chars`	Allowed: A–Z, a–z, 0–9, hyphen

Note: The schema FK on `SPECCODE` references `SPECCODE(Value)` as a hard stop.

TAXCODE

Check	Severity	Rule ID	Notes
TAXCODE value not in 0–9	error	`species_rules.taxcode_out_of_range`	Valid codes per `config.valid_taxcodes` (default 0–9)
TAXCODE missing on a sighting record	error	`species_rules.taxcode_missing_for_sighting`	Only applies when `config.require_taxcode_for_sightings` is true
TAXCODE not found in TAXCODE.csv lookup	warning	`species_rules.taxcode_not_in_table`	Soft check since 0–9 range is also enforced

SPECCODE/TAXCODE cross-check

Check	Severity	Rule ID	Notes
TAXCODE does not match the TAXCODE associated with SPECCODE in the lookup table	error	`species_rules.speccode_taxcode_mismatch`	Requires `TAXCODE` column in SPECCODE.csv

NUMBER (group size)

Check	Severity	Rule ID	Notes
NUMBER is negative	error	`species_rules.number_negative`	—
NUMBER is 0 on a sighting record	warning	`species_rules.number_zero_on_sighting`	Sighting with no animals is suspicious
NUMBER exceeds threshold	warning	`species_rules.number_unusual`	Data-driven SPECCODE → TAXCODE → global-default cascade (`typical_max_group`, default 1000); see `docs/validation_rules_guide.md`. Not a fixed number, and not a two-tier error/warning split — one warning-level check.
NUMBER is not an integer	error	`species_rules.number_not_integer`	Fractional counts are not valid

NUMCALF (calf count)

Check	Severity	Rule ID	Notes
NUMCALF is negative	error	`species_rules.numcalf_negative`	—
NUMCALF exceeds threshold	warning	`species_rules.numcalf_unusual`	Same SPECCODE → TAXCODE → global-default cascade as NUMBER (`typical_max_calf`, default 100)
NUMCALF > 0 for a non-mammal taxcode	warning	`species_rules.numcalf_non_mammal`	Calves only valid for marine mammals (TAXCODE 1–4)
NUMCALF is not an integer	error	`species_rules.numcalf_not_integer`	—
NUMCALF exceeds total NUMBER	error	`species_rules.numcalf_exceeds_total`	Calves cannot outnumber the group
NUMCALF exceeds half of NUMBER	warning	`species_rules.numcalf_exceeds_half`	Unusually calf-heavy group

Right whale specific

Check	Severity	Rule ID	Notes
Right whale group size unusually large	warning	`species_rules.right_whale_large_group`	Applies to RIWH, NARW, SARW; default: > 50
Right whale calf count unusually high	warning	`species_rules.right_whale_high_calf_count`	Default: > 5

2.9 **`platform_rules.m`** — Platform code validation

Rule ID prefix: `platform_rules`

Validates PLATFORM against the `PLATFORM.csv` lookup table. Redundant with the schema FK (FK_Master_PLATFORM) but catches the problem before upload with a descriptive message listing the bad values.

Check	Severity	Rule ID
PLATFORM value not in PLATFORM lookup table	error	`platform_rules.platform_invalid`

2.10 **`photos_rules.m`** — Photo documentation validation

Rule ID prefix: `photos_rules`

Validates PHOTOS against the `photos` lookup table (values 1–5). Redundant with the schema FK (FK_Master_PHOTOS).

Check	Severity	Rule ID
PHOTOS value not in photos lookup table	error	`photos_rules.photos_invalid`

2.11 **`temportal_rules.m`** — Temporal sequence rules (stub)

This file exists but contains no implemented checks. It is a placeholder for row-to-row time-sequence checks (times out of order, gaps between position records) that have not yet been implemented.

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3. SAS Legacy Coverage Mapping

The legacy SAS quality-control system comprised 11 check files. The table below lists every check from each SAS file and maps it to the corresponding enforcement in the new system (schema constraint, MATLAB rule, or gap).

Legend: Schema = enforced by SQL Server FK or NOT NULL constraint; MATLAB = enforced by a MATLAB validation rule; Gap = not yet implemented.

ChkBasic.sas — Universal checks (all survey types)

SAS check	New system coverage
FILEID missing	Schema — `FILEID NOT NULL`
EVENTNO missing	Schema — `EVENTNO NOT NULL`

SAS check	New system coverage
YEAR missing	MATLAB — `required_fields` (in configured field list)
YEAR/FILEID cross-consistency (derives year from FILEID prefix)	Gap — not implemented
MONTH missing	MATLAB — `required_fields`; also Schema FK to MONTH table
LATDEG/LATMIN missing (degree+minute form)	MATLAB — `coordinate_rules` checks LAT_DD missing; degree/minute components not validated separately
LONGDEG/LONGMIN missing (degree+minute form)	Same as above for LONG_DD
BEAUFORT > 7 (9 = missing sentinel)	Partial gap — `beaufort_rules` validates against Beaufort lookup; SAS treated 8 as an error and 9 as a “missing” sentinel; current MATLAB/schema allow 8 and do not interpret 9 as missing
CLOUD out of range (SAS valid: 0–4 and 9)	Schema FK to Cloud table (pending CLOUD scale resolution); `foreign_key_rules` checks against Cloud.csv
DAY out of range (0, > 31, calendar invalids)	MATLAB — `datetime_rules.day_out_of_range` (bounds) + `datetime_rules.invalid_date_combination` (calendar)
HEADING > 359	Gap — not implemented in any MATLAB rule
LATDEG out of range (25–48)	MATLAB — `coordinate_rules.outside_survey_lat` (warning) checks 20–55; exact SAS bounds (25–48) differ
LATMIN ≥ 60	Gap — MATLAB validates only decimal-degree LAT_DD; minute components not checked
LONGDEG out of range (58–81)	MATLAB — `coordinate_rules.outside_survey_lon` (warning) checks −85 to −40; exact SAS bounds differ
LONGMIN ≥ 60	Gap — same as LATMIN
MONTH > 12	MATLAB — `datetime_rules.month_out_of_range` (1–16); Schema FK to MONTH table
SURFTEMP out of range (−2 to 35 °C)	MATLAB — `environmental_rules` warnings at −2/35 °C (matches SAS thresholds)

SAS check	New system coverage
TIME missing except for opportunistic surveys	Gap — `datetime_rules` checks TIME format but does not implement the FILEID-based opportunistic exception
TIME ≥ 240000 or < 0	MATLAB — `datetime_rules.time_out_of_range`
TIME minutes ≥ 60	MATLAB — `datetime_rules.time_invalid_minutes`
TIME seconds ≥ 60	MATLAB — `datetime_rules.time_invalid_seconds`
WX invalid code	Schema FK to WX table; MATLAB — `foreign_key_rules.wx_invalid`
YEAR < 1990 or > 2018	MATLAB — `datetime_rules.year_out_of_range` (range 1900 to current+1); `datetime_rules.year_too_old` warning before 1980; upper bound is now dynamic rather than 2018

ChkBehav.sas — Behavior code validation

SAS check	New system coverage
Discontinued behavior code in BEHAV1–15 (codes 27, 29, 46, 47, 48, 64, 66)	Schema FK to Behave table + MATLAB `behavioral_rules.invalid_behavior_code` — catches them if they are absent from Behave.csv; the “discontinued” vs. “unknown” distinction is not preserved
Unknown behavior code in BEHAV1–10, BEHAV12–15 (codes 31–33, 39, 49, 56, 57, 96, 99)	Same as above
Unknown code in BEHAV11 only (additionally 71–75)	Gap — the BEHAV11-specific restriction on codes 71–75 is not implemented

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ChkCCSA.sas — CCS aerial survey checks

SAS check	New system coverage
GLARE > 3	Schema FK to GLARE table; MATLAB `foreign_key_rules.glarel_invalid` / `foreign_key_rules.glarer_invalid`
BLOCK code invalid	Schema FK to Block table; MATLAB `foreign_key_rules.block_invalid`
PORTOBS/STAROBS/SIGHTOBS observer codes	Gap — these fields are not in the 55-column MATLAB schema
STRATUM code invalid	Schema FK to STRATUM table; MATLAB `foreign_key_rules.stratum_invalid`
STRATUM on cross-leg or transit leg	Gap — survey-type cross-field check not implemented
STRIP missing when required	Gap — conditional required-field check not implemented
STRIP out of range (> 16, = 0, > 14 for non-626)	Schema FK to STRIP table; MATLAB `foreign_key_rules.strip_invalid`; platform-specific upper bound (> 14 for non-626) is a gap
Time format (invalid minutes or seconds)	MATLAB — `datetime_rules.time_invalid_minutes` / `datetime_rules.time_invalid_seconds`
Times out of order (row-to-row)	Gap — `temportal_rules.m` is a stub; not implemented
> 2 minutes without a position	Gap — not implemented
Duplicate location and time	Gap — not implemented
Same location, different time	Gap — not implemented
Speed too high (> 225 knots)	Gap — not implemented
Speed too low (0.5–50 knots)	Gap — not implemented
ALT missing	Gap — not in configured required fields for aerial surveys
ALT out of range (60–750 ft)	Gap — not implemented
ALT = 999 (bogus sentinel)	Gap — not implemented
Climb rate by platform	Gap — not implemented
Descent rate by platform	Gap — not implemented
Required fields missing (DAY, TIME, LEGTYPE, BEAUFORT, CLOUD, WX, VISIBLTY, HEADING, LEGSTAGE, LEGNO, STRATUM, GLAREL, GLARER)	Partial — `required_fields` checks what is in the configured list; not all of these are currently configured as required

SAS check	New system coverage
LEGTYPE out of range (valid: 1–4, 7)	Schema FK to LEGTYPE table; MATLAB `foreign_key_rules.legtype_invalid`; survey-type range restriction (CCS only 1–4, 7) is a gap
LEGTYPE=4 without preceding LEGSTAGE=3	Gap — state machine check not implemented
LEGSTAGE sequence state machine	Gap — not implemented
Last record LEGSTAGE ≠ 5	Gap — not implemented
PLATFORM out of range (626–649)	Schema FK to PLATFORM table; MATLAB `platform_rules`; survey-type range check (626–649 for CCS) is a gap
LEGSTAGE=7 with PHOTOS=1	Gap — not implemented
SIGHTNO ≥ 300 but LEGSTAGE ≠ 7	Gap — not implemented
SIGHTNO ≥ 300 but PHOTOS=1	Gap — not implemented

ChkDair.sas — Daily aerial survey checks

All checks from ChkCCSA apply (see above), plus:

SAS check	New system coverage
> 10 minutes without a position	Gap — not implemented (different threshold than ChkCCSA)
LEGSTAGE=7 extended state machine	Gap — not implemented
LEG STAGE on non-census leg	Gap — not implemented

ChkDupes.sas — Duplicate EVENTNO field consistency

SAS check	New system coverage
Repeated EVENTNO within a file where metadata fields differ across rows (TIME, MONTH, DAY, YEAR, coordinates, leg fields, environmental fields)	Gap — not implemented. `remove_duplicates.m` is marked do-not-use; no rule checks for metadata-inconsistent duplicate ENTRYNOs

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CHKDUPE2.sas — Duplicate SIGHTNO detection

SAS check	New system coverage
Duplicate SIGHTNO values within the same file (consecutive same value)	Gap — not implemented

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ChkIShip.sas — Intermittent shipborne survey checks

SAS check	New system coverage
MONTH/DAY change across consecutive records	Gap — not implemented
Time format (minutes, seconds out of range)	MATLAB — `datetime_rules.time_invalid_minutes` / `datetime_rules.time_invalid_seconds`
Times out of order (cross-day aware)	Gap — not implemented
ALT > 0 for ship survey (vessel airborne check)	Gap — not implemented
DAY, TIME, LEGTYPE missing	Partial — `required_fields` checks configured fields; not all are configured
BEAUFORT, CLOUD, VISIBLTY missing when LEGSTAGE present	Gap — conditional required-field check not implemented
LEGSTAGE missing after active stage	Gap — not implemented
LEGTYPE not 5 or 6 for intermittent ship	Schema FK to LEGTYPE table catches invalid codes; survey-type restriction to 5/6 is a gap
LEGSTAGE sequence (valid: 1, 2, 5)	Gap — not implemented

SAS check	New system coverage
Speed > 25 knots	Gap — not implemented

ChkPair.sas — Paired aerial survey checks

SAS check	New system coverage
Standard aerial checks (GLARE, STRATUM, STRIP, time format, times out of order, duplicate, speed, altitude, climb/descent rates)	Gap — see ChkCCSA gap entries above
ALT = 999 bogus sentinel	Gap — not implemented
LEGTYPE ≠ 7 and ≠ 9	Schema FK catches invalid codes; restriction to 7/9 for paired surveys is a gap
LEGSTAGE sequence (valid: 1, 2, 5)	Gap — not implemented
> 15 minutes without a position (on-effort only)	Gap — not implemented
Speed > 200 (non-654) or > 130 (platform 654)	Gap — not implemented
Speed < 25	Gap — not implemented
SST gradient > 0.5 °C/nm with SSTDIFF > 1 °C	Gap — not implemented; no equivalent anywhere in MATLAB
PLATFORM 600–699 range	Schema FK to PLATFORM table; paired-specific range check is a gap

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ChkShip.sas — Systematic shipborne survey checks

SAS check	New system coverage
MONTH/DAY change across consecutive records	Gap — not implemented
Time format and times out of order (cross-day)	Partial / Gap — time format checked by `datetime_rules`; sequential check not implemented
ALT > 0 for ship survey	Gap — not implemented

SAS check	New system coverage
DAY, TIME, LEGTYPE missing	Partial — `required_fields` covers configured fields
VISIBLTY > 20	MATLAB — `environmental_rules.visibility_too_high` warns at > 50 (looser than SAS threshold of > 20 for ships)
BEAUFORT, CLOUD, WX, VISIBLTY missing when LEGSTAGE present	Gap — conditional required-field check not implemented
LEGSTAGE missing/sequence	Gap — not implemented
LEGTYPE not 5 or 6	Schema FK catches invalid codes; restriction is a gap
Speed > 20 knots	Gap — not implemented
> 30 minutes without a position (on-effort)	Gap — not implemented

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ChkSight.sas — Sighting field validation

SAS check	New system coverage
CONFIDNC missing	MATLAB — `required_fields` if configured; Schema FK allows NULL
CONFIDNC > 11	Schema FK to Confidnc table; MATLAB `foreign_key_rules.confidnc_invalid`
CONFIDNC too large for NUMBER (matrix: group size → max confidence)	Gap — not implemented
CONFIDNC = 1 for NUMBER = 1 (±1 is illogical for a single animal)	Gap — not implemented
CONFIDNC = 11 with non-missing NUMBER (no-count code with an actual count)	Gap — not implemented

SAS check	New system coverage
NUMBER > 20 with CONFIDNC=0 for non-RECV/SPFV species	Gap — not implemented
DEPTH > 2000 m	Gap — DEPTH field is not validated
IDREL missing	MATLAB — `required_fields` if configured
IDREL = 0 or 4–8	Schema FK to IDREL table; MATLAB `foreign_key_rules.idrel_invalid`
IDREL = 1 for unidentified species (SPECCODE starts with ‘UN’)	Gap — not implemented
NUMBER missing when CONFIDNC ≠ 11	Gap — conditional required-field check not implemented
NUMBER = 0	MATLAB — `species_rules.number_zero_for_sighting` (warning)
NUMBER too high by TAXCODE (per-taxcode matrix)	Partial gap — `species_rules.number_unusual` flags NUMBER exceeding the SPECCODE/TAXCODE/global-default cascade threshold (warning-only, data-driven, not a fixed number); per-taxcode thresholds from SAS (TAXCODE 1/2: > 30; TAXCODE 5: > 5; etc.) are not implemented
NUMCALF ≥ 5	Partial — `species_rules.right_whale_high_calf_count` (warning at > 5 for right whales); generic `species_rules.numcalf_unusual` triggers at the SPECCODE/TAXCODE/global-default cascade threshold for all species
NUMCALF not less than NUMBER	MATLAB — `species_rules.numcalf_exceeds_total` (error)
PHOTOS missing	MATLAB — `required_fields` if configured
PHOTOS > 5	Schema FK to PHOTOS table; MATLAB `foreign_key_rules.photos_invalid` and `photos_rules.photos_invalid`

ChkSpeci.sas — SPECCODE and TAXCODE validation

SAS check	New system coverage
SPECCODE missing	MATLAB — `species_rules.speccode_missing_for_sighting`
SPECCODE not in hardcoded 2015 species list	Schema FK to SPECCODE table; MATLAB `species_rules.speccode_not_in_table` — lookup-table approach is more maintainable than the hardcoded SAS list
TAXCODE missing	MATLAB — `species_rules.taxcode_missing_for_sighting`

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4. Open Issues Affecting Coverage

1. CLOUD scale — ChkBasic.sas treats only 0–4 and 9 as valid; Cloud.csv currently covers 0–9. Domain expert review required to determine correct valid set before either approach can be declared authoritative.
2. HEADING range — ChkBasic flags HEADING > 359 as an error. No MATLAB rule or schema constraint enforces this. Should be added to an existing rule (e.g., `environmental_rules.m` or a new `platform_rules` check).
3. YEAR/FILEID mismatch — ChkBasic derives an expected year from the FILEID prefix character and digits and flags a mismatch. This cross-field check has no MATLAB equivalent.
4. PORTOBS/STAROBS/SIGHTOBS — ChkCCSA validates these observer code fields. They are not present in the 55-column MATLAB schema; confirm whether they should be added.
5. Row-to-row temporal and spatial continuity — ChkCCSA, ChkDair, ChkPair, ChkShip, and ChkIShip all rely on comparing consecutive rows (times out of order, position gaps, speed). These require a different computation pattern (sorted, grouped iteration) and have no equivalent anywhere in the current MATLAB rules. They represent the largest class of unported checks.
6. LEGSTAGE state machine — ChkCCSA and ChkDair validate that LEGSTAGE transitions follow a legal sequence within a file. This is a row-to-row check and is also not implemented.
7. Conditional required fields — Several SAS checks require fields only when certain conditions are met (ALT required for aerial, BEAUFORT/CLOUD/VISIBLTY required when LEGSTAGE is non-missing, TIME required except for opportunistic surveys). The `required_fields` rule is not yet parameterized for these conditions.

8. CONFIDNC/NUMBER matrix — ChkSight.sas validates that the confidence level is appropriate for the group size (e.g., CONFIDNC cannot be 8 for a group of 1). This cross-field check is not implemented.
9. Duplicate SIGHTNO and metadata-inconsistent duplicate EVENTNO — ChkDupes.sas and CHKDUPE2.sas are entirely unimplemented in the MATLAB system.
10. BEHAV11 asymmetry — ChkBehav.sas applies an extra restriction on codes 71–75 specifically for the BEHAV11 slot. This is not implemented and may or may not be intentional domain logic.